

Master Software Test Plan

IEEE 829 Standard Compliant Verification, Validation & Automated Testing Strategy

Production documentation artifact for the AI-Powered Household Appliance Rental System (RentAI / SDC2), encompassing enterprise Pure NoSQL architecture, customer churn prevention, multi-paradigm collaborative filtering, and local LLM context assistance.

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Project Title:	RentAI / SDC2 Household Appliance Ecosystem	Classification:	Confidential / Academic & Enterprise
Lead Architecture:	Deep Learning & Full-Stack Systems Team	Publication Date:	September 2026
Status:	VERIFIED & APPROVED	Target Platform:	Ubuntu 24.04 LTS / Windows 11 Enterprise

1. Introduction & Objectives

This Master Test Plan establishes the verification framework for RentAI / SDC2. Testing objectives include validating data consistency across MongoDB collections, guaranteeing $\geq 95\%$ accuracy in LightGBM churn predictions, verifying sub-50ms API throughput, and validating front-end component responsiveness across client viewports.

2. Testing Levels & Strategy

Test Level	Scope	Target Tool	Responsibility
Unit Testing	MongoEngine validation, feature transforms	PyTest / Unittest	Backend Engineer
Integration Testing	API endpoints, JWT lifecycle, DB writes	Django Test Client	QA Engineer
ML Validation	K-fold cross-validation, ROC-AUC, RMSE	Scikit-Learn metrics	ML Scientist
System & E2E	Full booking flow, RAG chat streaming	Manual & Automated scripts	Lead QA

3. Test Traceability Matrix

Requirement ID	Description	Test Case ID	Success Criteria
FR-CAT-01	Catalog filtering by price & category	TC-API-CAT-01	HTTP 200; all items match filter criteria
FR-CHURN-02	LightGBM Churn Scoring	TC-ML-CHURN-01	ROC-AUC ≥ 0.95 ; latency $\leq 20\text{ms}$
FR-REC-03	FunkSVD Top-K Recommendations	TC-ML-REC-01	RMSE ≤ 1.30 ; 100% catalog coverage
FR-CHAT-04	Ollama Auto-Detection & Inference	TC-AI-CHAT-01	Auto-starts if down; HTTP 200 response